11259A283.1.py

```python
import numpy as np
a = np.arange( start_or_stop: 9, dtype=np.float64).reshape(3, 3)
print("First Array:")
print(a)
b = np.array([10, 20, 30])
print("Second Array:")
print(b)
print("Addition of Arrays:")
print(np.add(a, b))
print("Subtraction of Arrays:")
print(np.subtract(a, b))
print("Multiplication of Arrays:")
print(np.multiply(a, b))
print("Division of Arrays:")
print(np.divide(a, b))
```

Run 11259A283.1

```
 [ -4. -13. -22.]]
Multiplication of Arrays:
[[  0.  20.  60.]
 [ 30.  80. 150.]
 [ 60. 140. 240.]]
Division of Arrays:
[[0.         0.05       0.06666667]
 [0.3        0.2        0.16666667]
 [0.6        0.35       0.26666667]]

Process finished with exit code 0
```

PythonProject › 11259A283.1.py 99:23 CRLF UTF-8 4 spaces Types: Built-in .venv [3.12.10]

11259A283.2.py 11259A283.3.py

```
83
84  units = 250
85  if units <= 100:
86      bill = units * 2
87  elif units <= 200:
88      bill = 100 * 2 + (units - 100) * 3
89  else:
90      bill = 100 * 2 + 100 * 3 + (units - 200) * 5
91  print("Electricity Bill:", bill)
92
93  expenses = [12000, 15000, 18000, 11000, 17000]
94  highest = max(expenses)
95  month = expenses.index(highest) + 1
96  print("Highest Expense:", highest)
97  print("Month:", month)
98
99
100
```

Run 11259A283.3

```
25 Adult
45 Adult
67 Senior Citizen
15 Child
Electricity Bill: 750
Highest Expense: 18000
Month: 3

Process finished with exit code 0
```

PythonProject4 Version Control Current File Trial

11259A283.2.py 11259A283.3.py 11259A283.4.py 11259A283.5.py

```python
# Dataset Splitting and Cross-Validation
# Using Iris Dataset

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split, KFold, cross_val_score
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

# ------------------------------------------------
# Step 1: Load the Iris Dataset
# ------------------------------------------------

iris = load_iris()

X = iris.data
y = iris.target

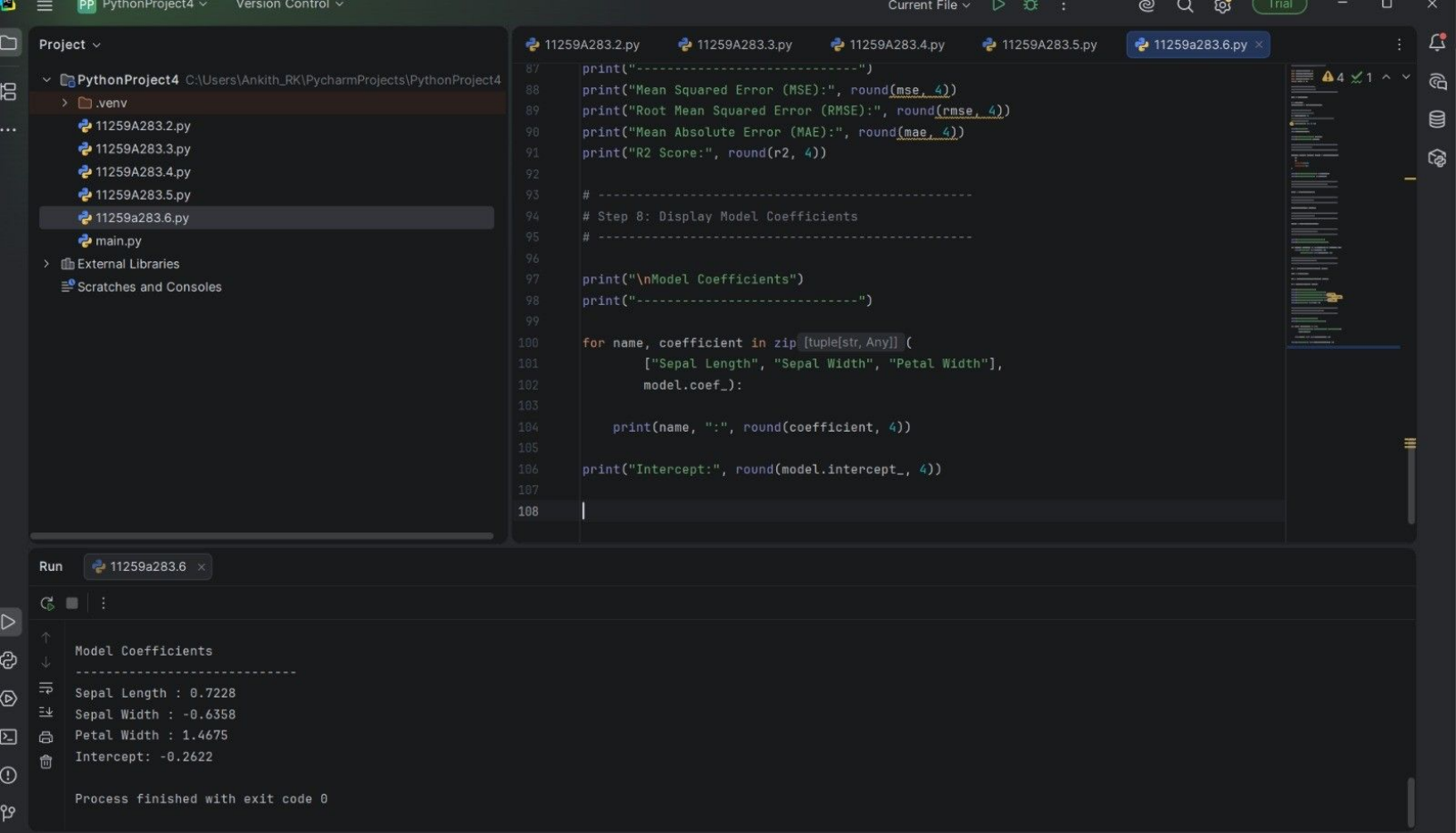
print("Dataset Shape:", X.shape)
print("Number of Samples:", len(X))
print("Number of Features:", X.shape[1])
print("Target Classes:", iris.target_names)

# ------------------------------------------------
```

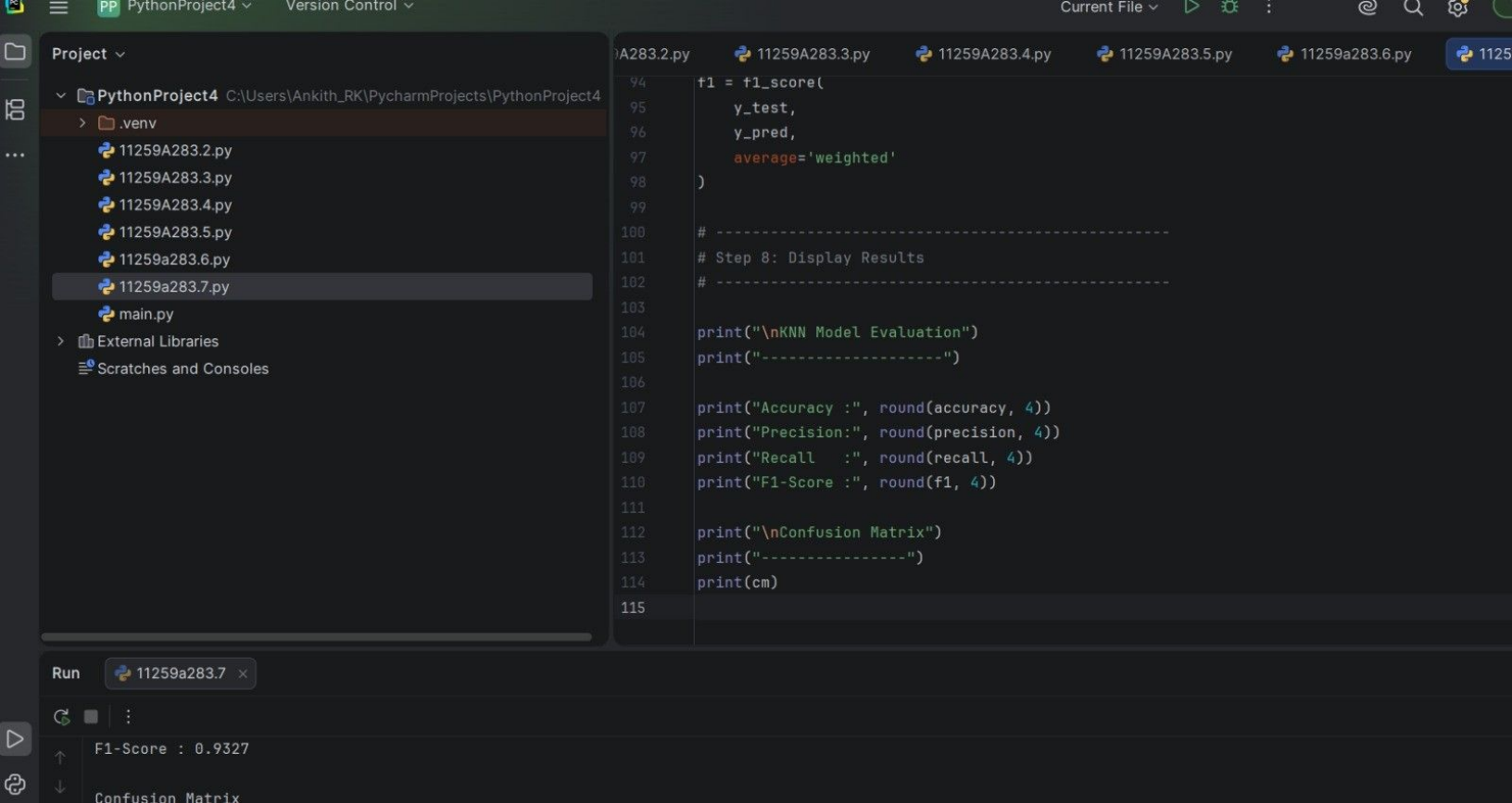
Run 11259A283.5

```
Fold 4 Accuracy: 0.9667
Fold 5 Accuracy: 0.9667

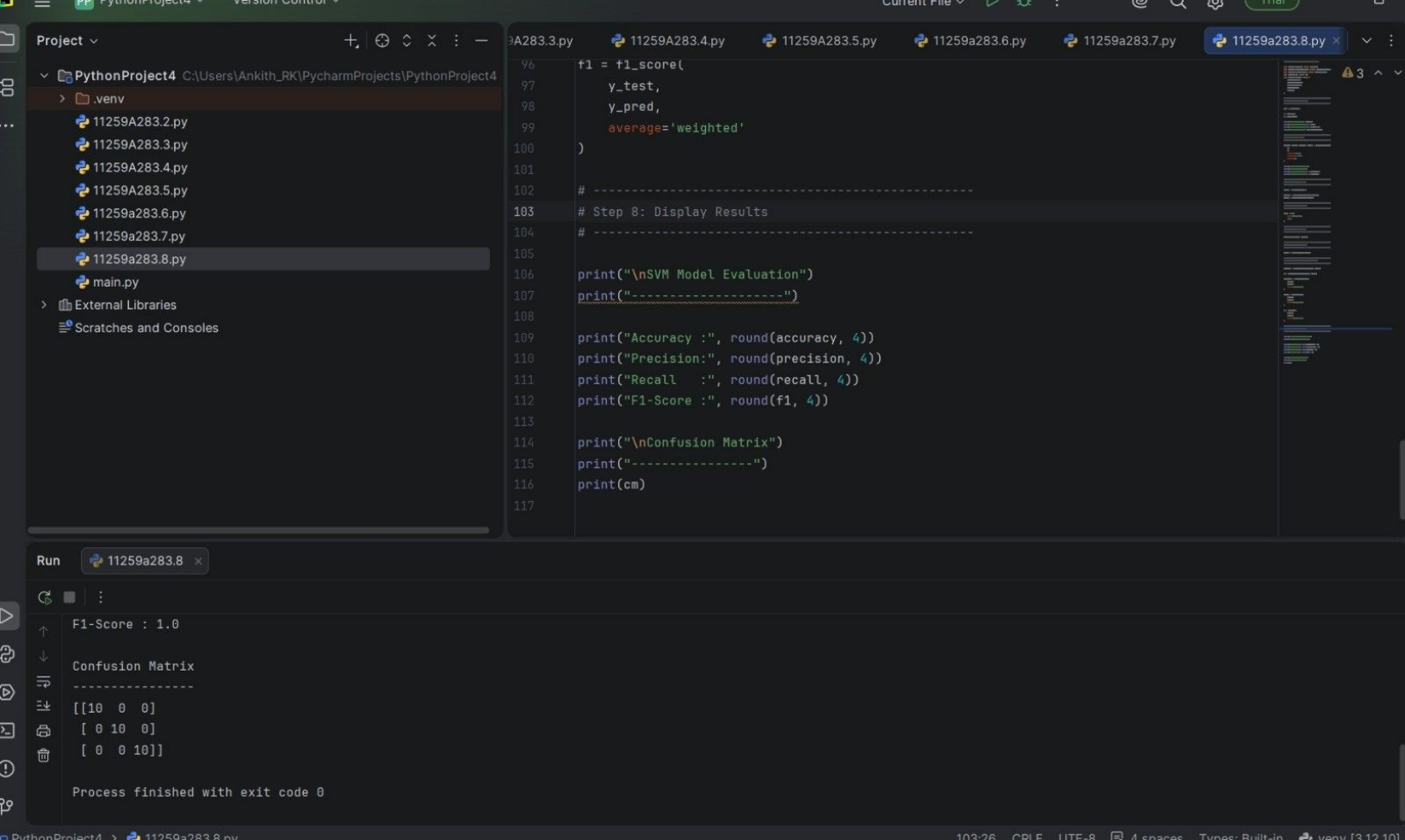
Cross-Validation Results
```


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11259A283.4.py
11259A283.5.py
11259a283.6.py
print("----------")
print("Mean Squared Error (MSE):", round(mse, 4))
print("Root Mean Squared Error (RMSE):", round(rmse, 4))
print("Mean Absolute Error (MAE):", round(mae, 4))
print("R2 Score:", round(r2, 4))
--
Step 8: Display Model Coefficients
--
print("\nModel Coefficients")
print("----------")
for name, coefficient in zip [tuple[str, Any]] (
["Sepal Length", "Sepal Width", "Petal Width"],
model.coef_):
print(name, ":", round(coefficient, 4))
print("Intercept:", round(model.intercept_, 4))
Run
11259a283.6
Model Coefficients

Sepal Length : 0.7228
Sepal Width : -0.6358
Petal Width : 1.4675
Intercept: -0.2622
Process finished with exit code 0


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11259a283.6.py
f1 = f1_score(
y_test,
y_pred,
average='weighted'
)

Step 8: Display Results
----
print("\nKNN Model Evaluation")
print("--------------------")
print("Accuracy :", round(accuracy, 4))
print("Precision:", round(precision, 4))
print("Recall :", round(recall, 4))
print("F1-Score :", round(f1, 4))
print("\nConfusion Matrix")
print("----------------")
print(cm)
Run
11259a283.7
F1-Score : 0.9327
Confusion Matrix


Project
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11259A283.4.py
11259A283.5.py
11259a283.6.py
11259a283.7.py
11259a283.8.py
f1 = f1_score(
y_test,
y_pred,
average='weighted'
)

Step 8: Display Results
----
print("\nSVM Model Evaluation")
print("--------------------")
print("Accuracy :", round(accuracy, 4))
print("Precision:", round(precision, 4))
print("Recall :", round(recall, 4))
print("F1-Score :", round(f1, 4))
print("\nConfusion Matrix")
print("----------")
print(cm)
Run
11259a283.8
F1-Score : 1.0
Confusion Matrix
[[10 0 0]
[0 10 0]
[0 0 10]]
Process finished with exit code 0

Project
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11259a283.8.py
11259a283.9.py
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11259a283.9.p
98
99 plt.figure(fig
100
101 plt.scatter(
102 X_scaled[:
103 X_scaled[:
104 c=labels
105)
106
107 plt.scatter(
108 centroids[
109 centroids[
110 marker='X'
111 s=200
112)
113
114 plt.xlabel("Pe
115 plt.ylabel("Pe
116 plt.title("K-M
117
118 plt.show()
119
Plots
1:1
800x600 PNG (32-bit color) 29.95 kB
K-Means Clustering on Iris Dataset
Petal Width
Petal Length
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
Run
11259a283.9

Silhouette Score: 0.4599
Number of Samples in Each Cluster:
Cluster 0 : 53 samples
Cluster 1 : 50 samples
Cluster 2 : 47 samples
Process finished with exit code 0

PythonProject4 | Version Control | Current File | Trial

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- 11259a283.6.py
- 11259a283.7.py
- 11259a283.8.py
- 11259a283.9.py
- 11259a283.10.py
- main.py

External Libraries

Scratches and Consoles

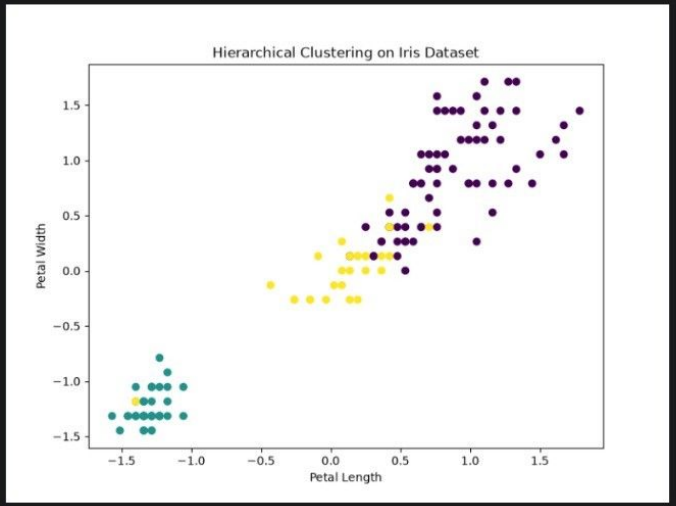
11259a283.10

Install Ignore

```
97
98
99   # ------------
100  # Step 8: Visu
101  # ------------
102
103  plt.figure(fig
104
105  plt.scatter(
106      X_scaled[:
107      X_scaled[:
108      c=labels
109  )
110
111  plt.xlabel("Pe
112  plt.ylabel("Pe
113  plt.title("Hie
114
115  plt.show()
116
```

Plots

800x600 PNG (32-bit color) 29.71 kB



Run 11259a283.10

```
Silhouette Score: 0.4467

Number of Samples in Each Cluster:
Cluster 0 : 71 samples
Cluster 1 : 49 samples
Cluster 2 : 30 samples

Process finished with exit code 0
```

PythonProject4 | Version Control | Current File | Trial

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- 11259a283.6.py
- 11259a283.7.py
- 11259a283.8.py
- 11259a283.9.py
- 11259a283.10.py
- 112592283.11.py
- main.py

External Libraries

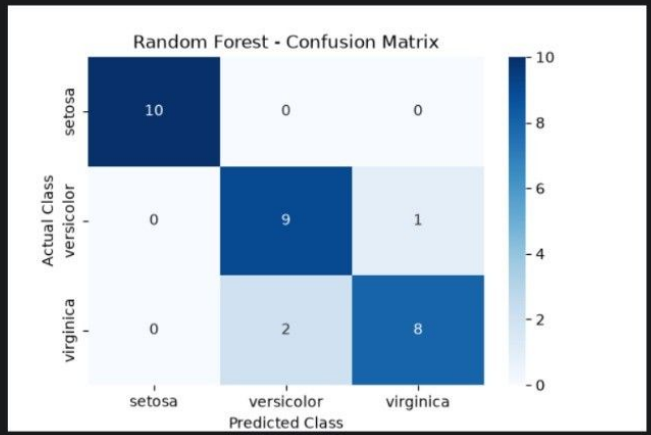
Scratches and Consoles

112592283.11

```
57  print(classif
58      y_test,
59      y_pred,
60      target_name
61  ))
62
63  # 9. Display Co
64  plt.figure(figs
65  sns.heatmap(
66      cm,
67      annot=True,
68      fmt="d",
69      cmap="Blues
70      xticklabels
71      yticklabels
72  )
73
74  plt.xlabel("Pre
75  plt.ylabel("Act
76  plt.title("Rand
77  plt.show()
78
```

Plots

600x400 PNG (32-bit color) 23.03 kB



Run 112592283.11

```
   versicolor       0.82      0.90      0.86        10
    virginica       0.89      0.80      0.84        10

     accuracy                           0.90        30
    macro avg       0.90      0.90      0.90        30
 weighted avg       0.90      0.90      0.90        30


Process finished with exit code 0
```

PythonProject4 | Version Control | Current File

Project

PythonProject4 C:\Users\Ankith_RK\PycharmProjects\PythonProject4

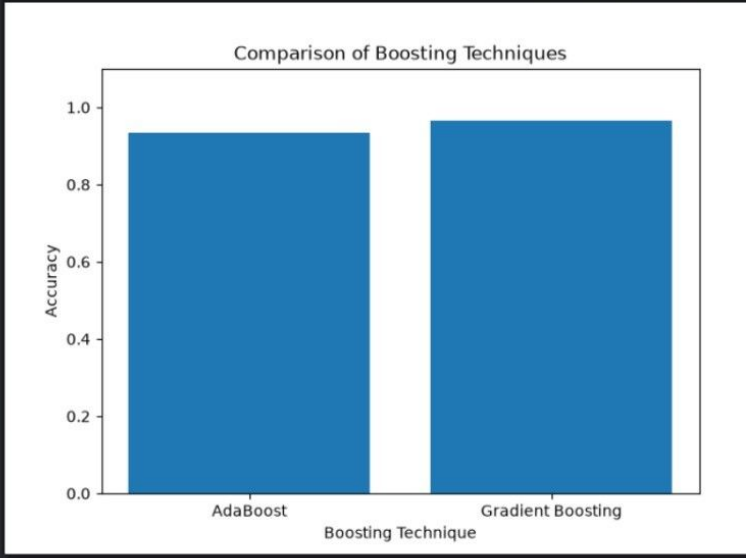
- .venv
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- 11259A283.4.py
- 11259A283.5.py
- 11259a283.6.py
- 11259a283.7.py
- 11259a283.8.py
- 11259a283.9.py
- 11259a283.10.py
- 11259A283.12.py
- 112592283.11.py
- main.py
- External Libraries
- Scratches and Consoles

11259A283.12

```
76
77      y_test,
78      gb_pred
79  )
80
81  # 10. Compare a
82  models = ["AdaB
83  accuracies = [
84      ada_results
85      gb_results[
86  ]
87
88  plt.figure(figs
89  plt.bar(models,
90
91  plt.xlabel("Boo
92  plt.ylabel("Acc
93  plt.title("Comp
94
95  plt.ylim(0, 1.1
96  plt.show()
97
```

Plots

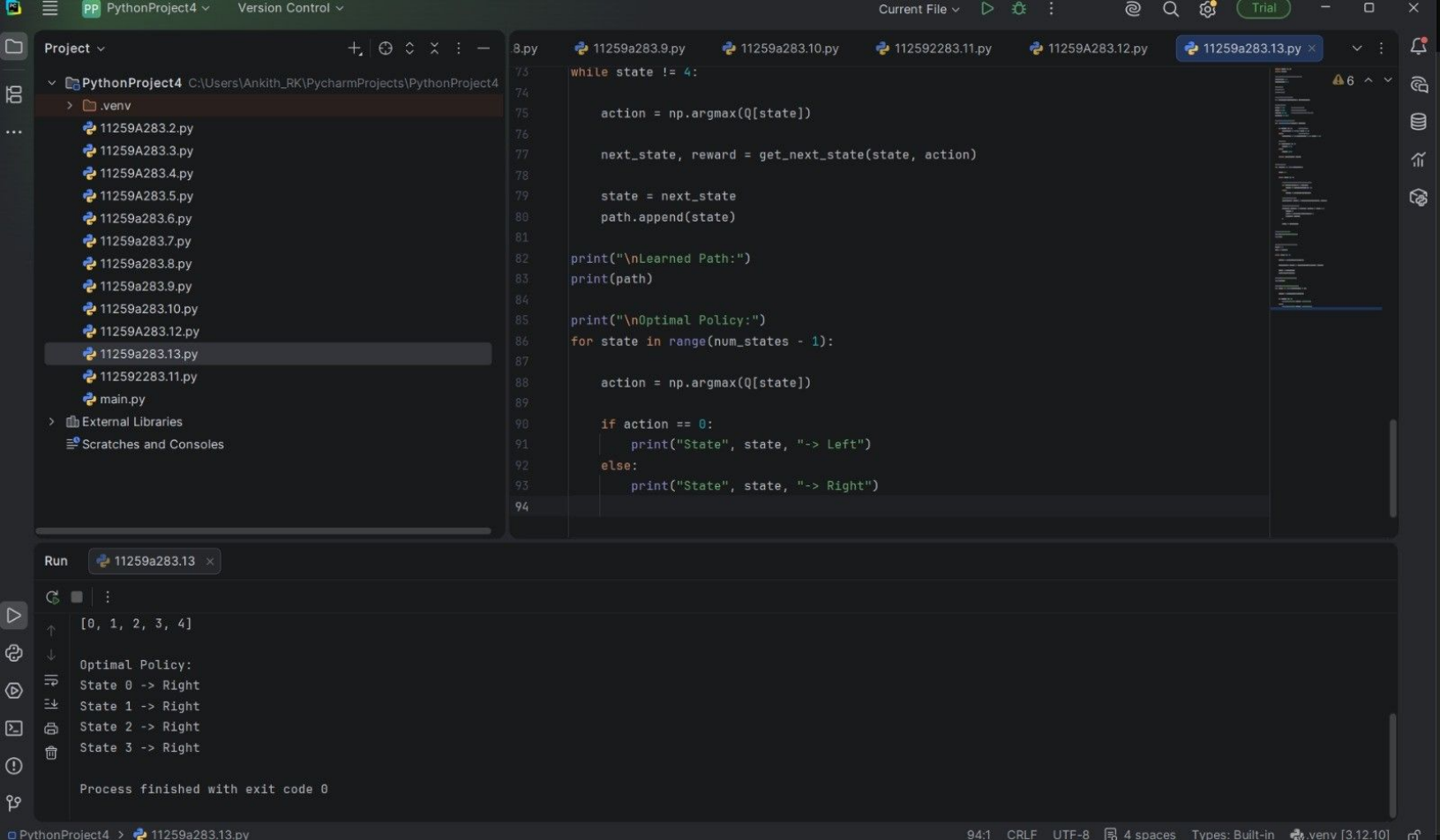
700x500 PNG (32-bit color) 2

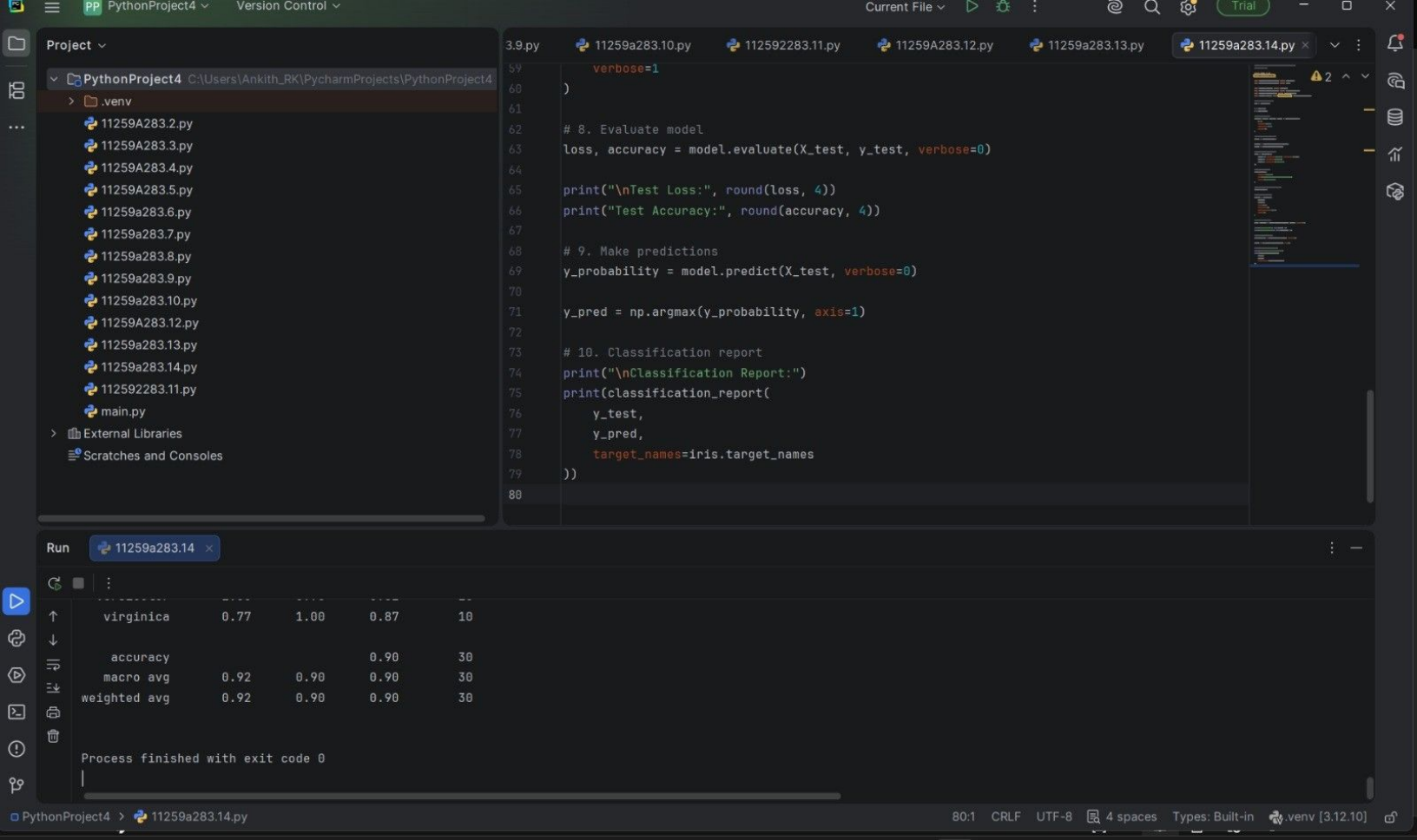


Run 11259A283.12

```
 Gradient Boosting
----------------------------
Accuracy : 0.9667
Precision: 0.9697
Recall   : 0.9667
F1-Score : 0.9666

Process finished with exit code 0
```


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11259a283.9.py
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112592283.11.py
11259A283.12.py
11259a283.13.py
while state != 4:
action = np.argmax(Q[state])
next_state, reward = get_next_state(state, action)
state = next_state
path.append(state)
print("\nLearned Path:")
print(path)
print("\nOptimal Policy:")
for state in range(num_states - 1):
action = np.argmax(Q[state])
if action == 0:
print("State", state, "-> Left")
else:
print("State", state, "-> Right")
Run
11259a283.13
[0, 1, 2, 3, 4]
Optimal Policy:
State 0 -> Right
State 1 -> Right
State 2 -> Right
State 3 -> Right
Process finished with exit code 0
PythonProject4 > 11259a283.13.py
94:1
CRLF
UTF-8
4 spaces
Types: Built-in
.venv [3.12.10]


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11259A283.12.py
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11259a283.14.py
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main.py
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3.9.py
11259a283.10.py
112592283.11.py
11259A283.12.py
11259a283.13.py
11259a283.14.py
verbose=1
)
8. Evaluate model
loss, accuracy = model.evaluate(X_test, y_test, verbose=0)
print("\nTest Loss:", round(loss, 4))
print("Test Accuracy:", round(accuracy, 4))
9. Make predictions
y_probability = model.predict(X_test, verbose=0)
y_pred = np.argmax(y_probability, axis=1)
10. Classification report
print("\nClassification Report:")
print(classification_report(
y_test,
y_pred,
target_names=iris.target_names
))
Run
11259a283.14
virginica 0.77 1.00 0.87 10
accuracy 0.90 30
macro avg 0.92 0.90 0.90 30
weighted avg 0.92 0.90 0.90 30
Process finished with exit code 0
PythonProject4 > 11259a283.14.py
80:1 CRLF UTF-8 4 spaces Types: Built-in .venv [3.12.10]